

1. Complete ICU Bed Booking Flowchart

flowchart



2. Patient Website Form

Section A — Request Type

- Booking request by Patient
- Booking request by Attendant
- Emergency / Planned admission
- Transfer from another hospital

Section B — Patient Details

- Patient full name
- Age / date of birth
- Gender
- Mobile number
- Alternate mobile number
- Address, city, state and PIN code
- Aadhaar/ID number — optional or policy-based
- Patient UHID — if already registered at NIMS

Section C — Attendant Details

यह section attendant द्वारा form भरने पर mandatory होगा।

- Attendant name
- Relationship with patient
- Mobile number
- Alternate number
- Email
- ID proof
- Same address as patient checkbox

Section D — Medical Information

- Required ICU type
- Current medical condition
- Diagnosis
- Symptoms/criticality
- Oxygen/ventilator requirement
- Treating doctor name
- Referring hospital
- Current hospital/location
- Expected admission date and time
- Ambulance required
- Infection/isolation requirement
- Additional remarks

ICU type options:

- Medical ICU
- Surgical ICU
- Cardiac ICU
- Neuro ICU
- Pediatric ICU
- Neonatal ICU
- Isolation ICU
- Ventilator bed

Section E — Documents

- Doctor referral letter
- Latest medical report
- Prescription
- Discharge/transfer summary
- Patient ID proof
- Attendant ID proof

Allowed formats: PDF, JPG, JPEG, PNG.

File size और file count backend से नियंत्रित हों।

Section F — Declaration

- दी गई जानकारी सही होने की सहमति
- Hospital verification की सहमति
- ICU bed केवल admin confirmation के बाद reserved होने की स्वीकृति
- Privacy/data-processing consent
- Emergency disclaimer
- Terms and conditions acceptance

3. Patient-Side Request Status

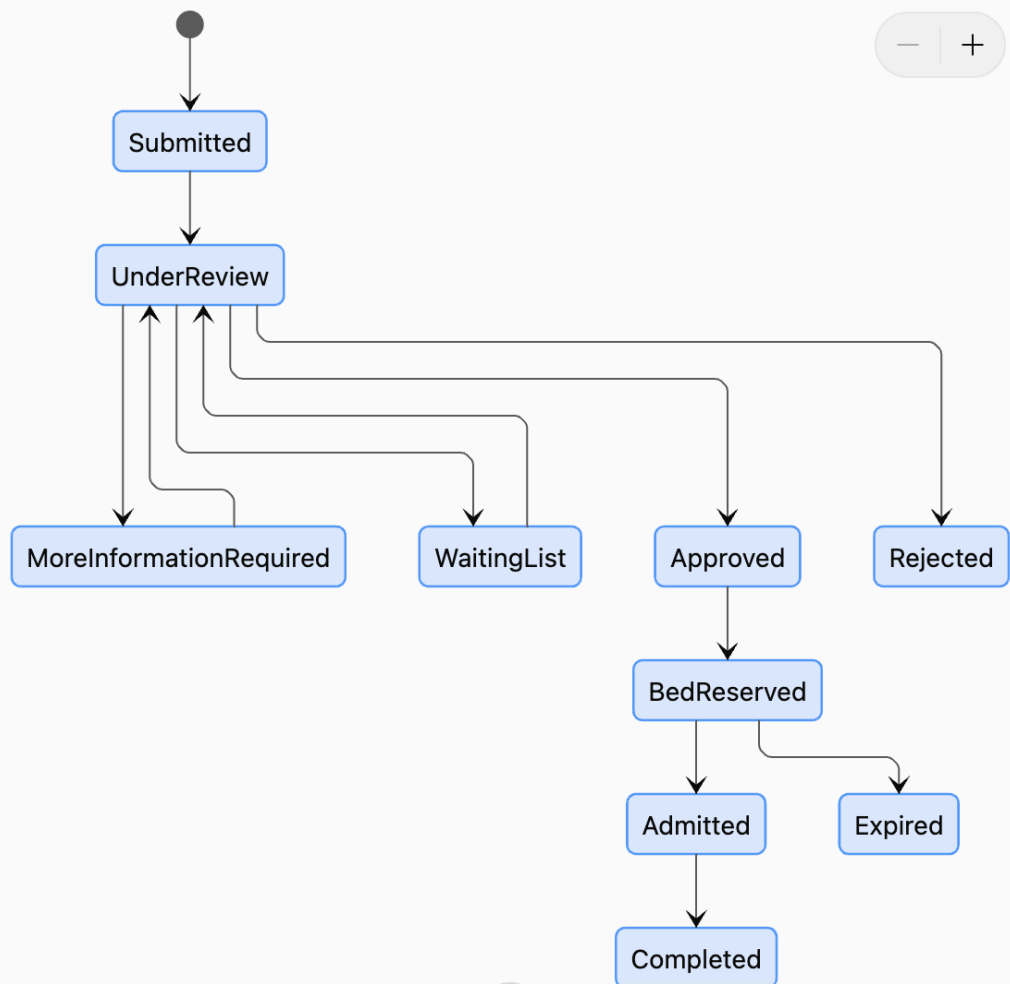
3. Patient-Side Request Status

Submit होने पर patient को एक unique ID मिले:

NIMS-ICU-20260912-0001



Status journey:



↓
Patient request ID और registered mobile number से status देख सके:

- Request submitted
- Under review
- More information required
- Waiting for bed
- Approved
- Bed temporarily reserved
- Admission completed
- Rejected
- Cancelled
- Reservation expired

4. Admin Panel Flow



5. Admin Dashboard Requirements

Dashboard cards:

- Total ICU beds
- Available beds
- Temporarily reserved beds
- Occupied beds
- Beds under maintenance
- New requests
- Critical requests
- Pending reviews
- Waiting-list patients
- Today's confirmed admissions

Admin filters:

- Request date
- ICU type
- Emergency/planned
- Request status
- Patient location
- Referring hospital
- Assigned doctor/admin
- Ventilator requirement
- Priority level

Admin actions:

- View complete request
- Preview/download reports
- Call patient/attendant
- Add internal notes
- Assign medical reviewer
- Request additional documents
- Set priority
- Add to waiting list
- Approve/reject request
- Select and temporarily reserve bed
- Set reservation expiry
- Mark patient arrived
- Complete admission
- Cancel or extend reservation
- View complete audit history

6. Recommended Roles

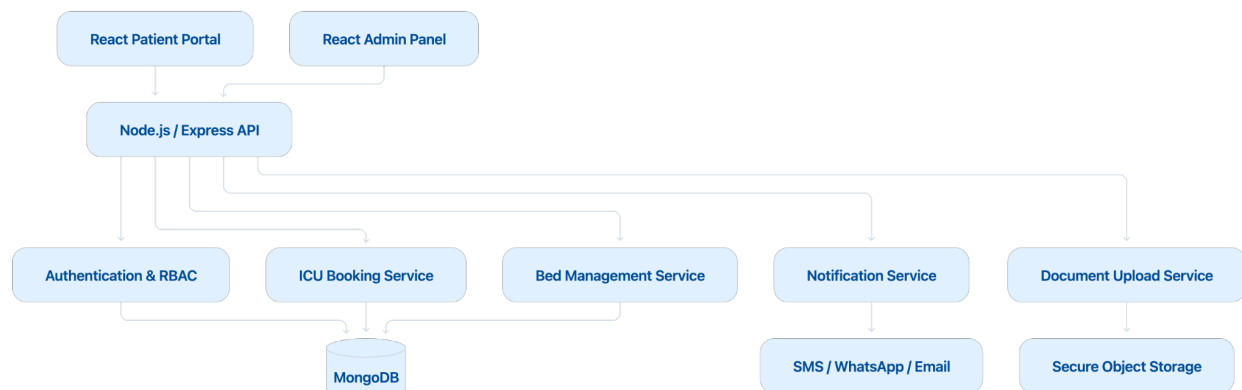
Role

Permissions

Super Admin	Complete system and configuration access
ICU Admin	Requests, beds, approvals and admissions
Doctor/Medical Reviewer	Medical information, reports and recommendation
Admission Desk	Patient verification and admission completion
Bed Manager	Bed availability and reservation management
Viewer/Management	Dashboard and reports only

7. Technical Architecture

flowchart TD



Recommended stack:

- Frontend: React.js
- UI: Material UI, Ant Design या Tailwind CSS
- Backend: Node.js with Express.js
- Database: MongoDB with Mongoose
- Authentication: JWT access token + refresh token
- OTP: SMS provider
- Documents: private S3-compatible object storage
- Notifications: SMS, WhatsApp and email
- Real-time admin alerts: Socket.IO
- Logging: Winston/Pino
- API documentation: Swagger/OpenAPI

8. Suggested MongoDB Collections

icu_requests

```
{
  requestId: "NIMS-ICU-20260912-0001",
  submittedBy: "attendant",

  patient: {
    name: "Patient Name",
    age: 58,
    gender: "male",
    mobile: "98XXXXXXXXX",
    uhid: null,
    address: {
      city: "Jaipur",
      state: "Rajasthan",
      pinCode: "302017"
    }
  },

  attendant: {
    name: "Attendant Name",
    relationship: "Son",
    mobile: "98XXXXXXXXX"
  },

  medical: {
    admissionType: "emergency",
    requiredIcuType: "medical_icu",
    diagnosis: "Medical condition",
    ventilatorRequired: true,
    oxygenRequired: true,
    referringDoctor: "Doctor Name",
    referringHospital: "Hospital Name"
  },

  documents: [
    {
      type: "medical_report",
      fileKey: "private/path/report.pdf"
    }
  ],

  priority: "critical",
  status: "under_review",
  assignedTo: null,

  reservation: {
```

```
    bedId: null,  
    reservedAt: null,  
    expiresAt: null  
  },  
  
  consentAccepted: true,  
  createdAt: new Date(),  
  updatedAt: new Date()  
}
```

Other collections:

- `users`
- `roles`
- `icu_units`
- `icu_beds`
- `bed_reservations`
- `request_status_history`
- `admin_notes`
- `notifications`
- `documents`
- `audit_logs`
- `otp_verifications`

9. Important API Endpoints

```
POST /api/icu-requests  
POST /api/icu-requests/send-otp  
POST /api/icu-requests/verify-otp  
GET /api/icu-requests/status/:requestId  
PATCH /api/icu-requests/:id/patient-response
```

```
POST /api/admin/auth/login  
GET /api/admin/icu-requests  
GET /api/admin/icu-requests/:id  
PATCH /api/admin/icu-requests/:id/status  
PATCH /api/admin/icu-requests/:id/assign  
POST /api/admin/icu-requests/:id/notes  
POST /api/admin/icu-requests/:id/request-information
```

```
GET /api/admin/icu-beds  
POST /api/admin/icu-beds  
PATCH /api/admin/icu-beds/:id  
POST /api/admin/icu-beds/:id/reserve  
POST /api/admin/icu-beds/:id/release  
POST /api/admin/icu-beds/:id/occupy
```

GET /api/admin/dashboard

GET /api/admin/reports

10. Critical Development Rules

- Bed availability और reservation को **MongoDB transaction/atomic operation** से handle करें, ताकि एक bed दो patients को assign न हो।
- Form submission को confirmed booking न माना जाए।
- Reservation की expiry time अनिवार्य हो।
- Every status change का audit log रखा जाए।
- Patient को admin की private notes दिखाई न दें।
- Medical documents public URL पर उपलब्ध न हों।
- OTP, rate limiting, CAPTCHA और server-side validation लगाएं।
- Sensitive information logs में print न करें।
- Data encryption in transit और at rest रखें।
- Admin accounts पर strong password और preferably MFA लागू करें।
- Role-based access लगाएं।
- Backup, restore testing और access logs maintain करें।
- Approval/rejection के साथ admin name, timestamp और reason save हो।
- Critical patient requests को केवल form queue पर निर्भर न रखें; emergency escalation/call workflow भी रखें।